

Practical No. 13

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Aim : Calculate size of the Project using Function Point Metric

Theory

Function Point Analysis (FPA) is a software sizing technique used to measure the functional size of a software system from the user's point of view. The five major function types are External Inputs (EI), External Outputs (EO), External Inquiries (EQ), Internal Logical Files (ILF), and External Interface Files (EIF).

Function Point Complexity Weights

Function Type	Low	Average	High
External Input (EI)	3	4	6
External Output (EO)	4	5	7
External Inquiry (EQ)	3	4	6
Internal Logical File (ILF)	7	10	15
External Interface File (EIF)	5	7	10

Given Function Count for Online Food Ordering System

Function Type	Low	Average	High	Weighted FP
External Inputs (EI)	4	3	1	$4 \times 3 + 3 \times 4 + 1 \times 6 = 30$
External Outputs (EO)	2	3	1	$2 \times 4 + 3 \times 5 + 1 \times 7 = 30$
External Inquiries (EQ)	2	2	1	$2 \times 3 + 2 \times 4 + 1 \times 6 = 20$
Internal Logical Files (ILF)	2	2	1	$2 \times 7 + 2 \times 10 + 1 \times 15 = 49$
External Interface Files (EIF)	1	1	0	$1 \times 5 + 1 \times 7 + 0 \times 10 = 12$
Total Unadjusted Function Points (UFP)				141

Step 1: Calculate Unadjusted Function Points (UFP)

UFP = Sum of weighted function counts

$$\text{UFP} = 30 + 30 + 20 + 49 + 12 = 141$$

Step 2: Calculate Value Adjustment Factor (VAF)

Assume the 14 General System Characteristics (GSCs) are rated on a scale of 0 to 5. For this practical, the total degree of influence (TDI) is assumed to be **42**.

$$\text{VAF} = 0.65 + (0.01 \times \text{TDI})$$

$$\text{VAF} = 0.65 + (0.01 \times 42) = 0.65 + 0.42 = 1.07$$

Step 3: Calculate Adjusted Function Points

$$\text{Adjusted FP} = \text{UFP} \times \text{VAF}$$

$$\text{Adjusted FP} = 141 \times 1.07 = 150.87 \approx 151 \text{ Function Points}$$

Result

Measure	Calculated Value
Unadjusted Function Points (UFP)	141
Value Adjustment Factor (VAF)	1.07
Adjusted Function Points	150.87 $\approx$ 151 FP

Conclusion

The estimated functional size of the Online Food Ordering System is approximately **151 Function Points**. Function Point Analysis helps estimate software size independently of programming language and can be used as a basis for effort, cost, and schedule estimation.